PROCGEN RL

Teaching an agent to play levels it has never seen

Reinforcement learning on a procedurally generated platformer, raced against its own past selves



[Your name] · [Course, date]

THE IDEA

A platformer the agent can’t memorise

01

Procedural levels

A new level every episode, built from a seed: gaps, stairs, spikes and floating platforms.

02

A learning agent

PPO learns from reward alone: move right, don’t die, reach the flag. No hand-written rules.

03

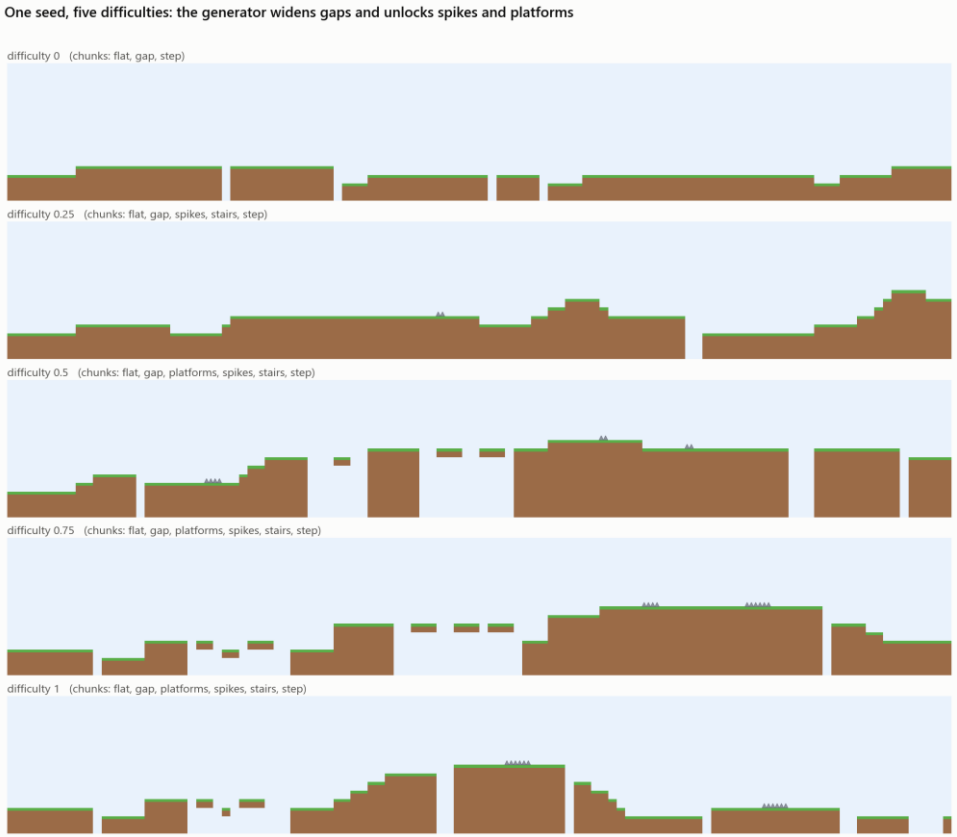
A ghost race

You play against snapshots of the agent after 0 min, 1 h, 10 h … 300 h of simulated training.

The question: does it learn **to play**, or does it just learn **these levels**?

THE GAME

Run right, reach the flag



Stage 1: terrain

Flat ground, steps, stairs and gaps, each sized from the jump physics.

Stage 2: hazards

Spikes from difficulty 0.25, floating platform hops from 0.35.

Physics

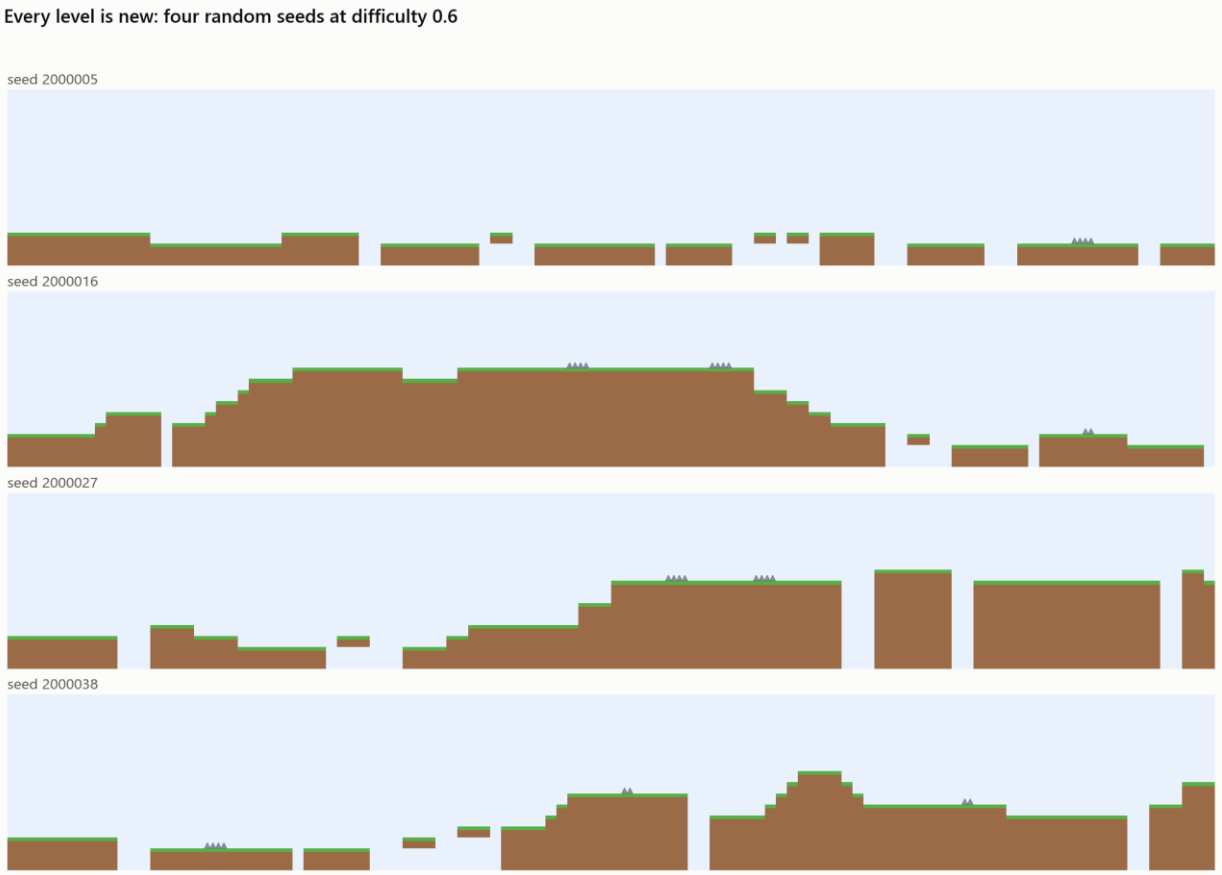
Deterministic 60 FPS tile physics: jumps up to 3.8 tiles, hold for higher, coyote time.

Difficulty 0 to 1 (demo: up to 2)

Widens gaps and height changes and adds hazards. Enemies are phase 2.

LEVEL GENERATOR

Every level is new and beatable



1,000,000

training seeds. A fresh level is drawn for every episode, built from chunks with a seeded random generator.

100%

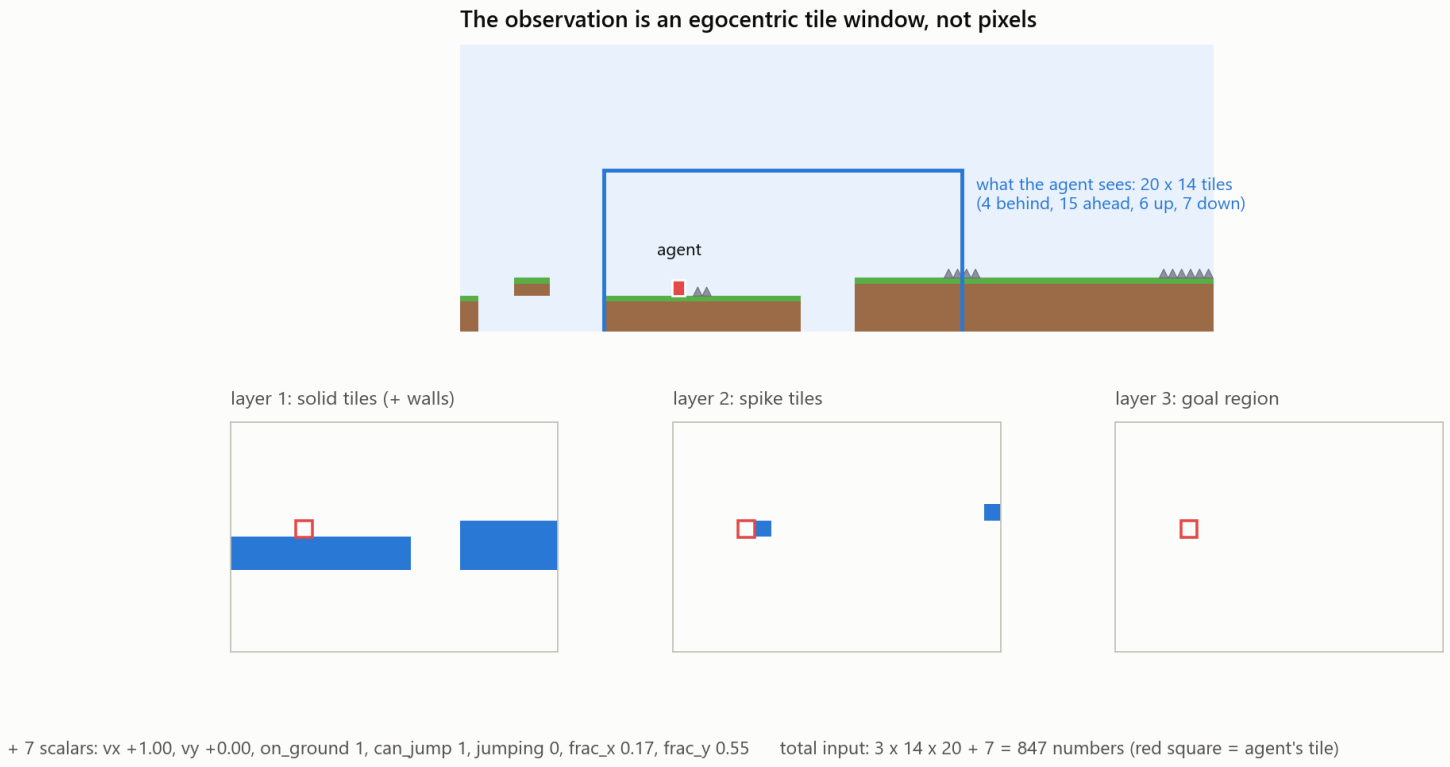
of tested levels solvable. A search-based solver beats every level in the test suite.

3 ranges

of seeds that never overlap: training, evaluation and the demo. The agent is never tested on a level it trained on.

OBSERVATION

What the agent sees



20 × 14 tiles

around the agent: 4 behind, 15 ahead

3 layers

solid, spike, goal; each tile is 0 or 1

+ 7 numbers

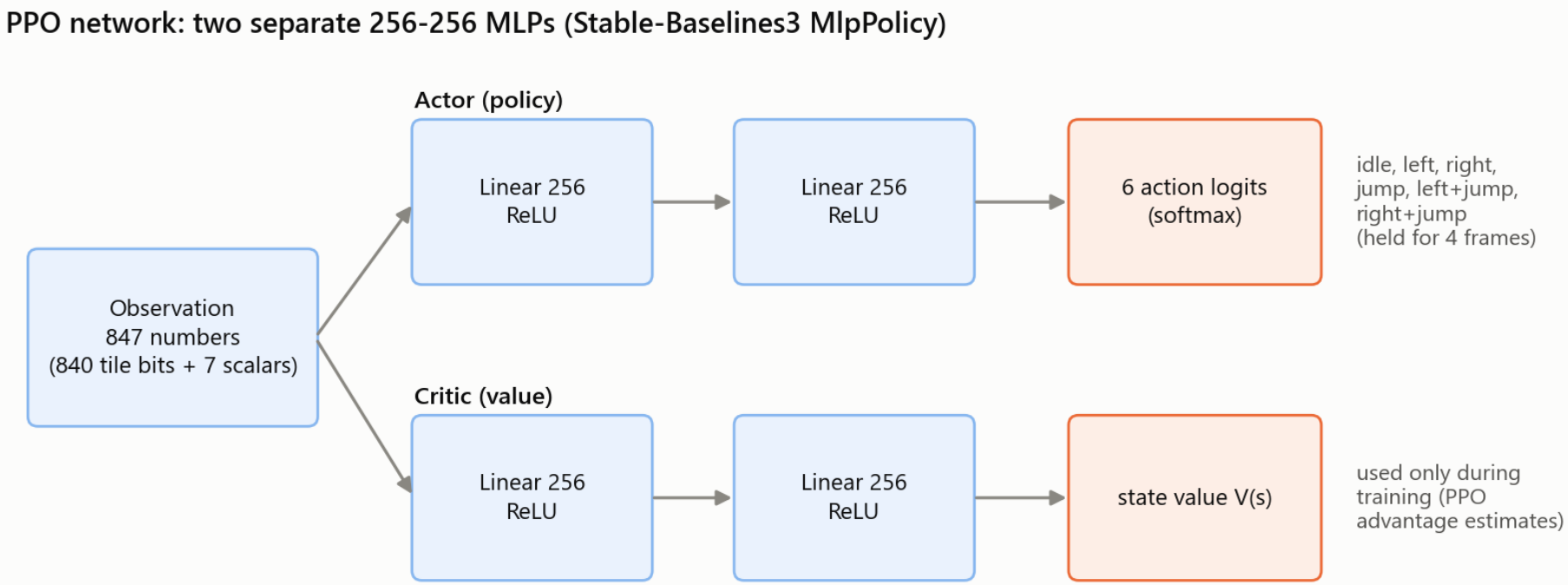
its own speed, on ground, can jump, jumping, position within the tile

847 inputs

cheaper than pixels, and the same on every level because it is relative to the agent

THE AGENT

PPO with two small networks



The actor picks one of 6 actions; the critic estimates how good the situation is. Only the actor ships in a ghost file, and the demo runs it with NumPy in about 0.1 ms.

REWARD AND TRAINING

Make progress, don’t die

Event	Reward
Each new tile of ground reached	+0.1
Each decision (time penalty)	−0.002
Falling into a pit or hitting spikes	−1
Reaching the flag	+5
No new ground for 90 decisions	episode ends

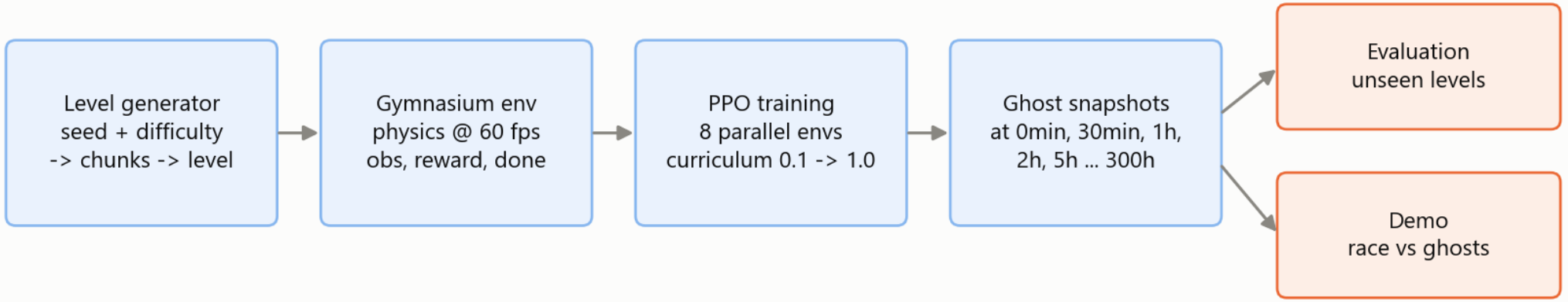
Training setup

- 6 actions: idle, left, right, jump, left + jump, right + jump
- One decision every 4 frames (15 per second)
- 8 games in parallel on a 16-thread laptop CPU
- About 2,700 decisions per second
- 1 simulated hour = 54,000 decisions

SYSTEM

From seed to ghost

System pipeline



Shared code path: physics.step_player + sim.Runner are used by training, evaluation and the demo, so a ghost behaves exactly as it did during training. Levels are deterministic per seed on every machine.

Runs in the web app on the laptop (or the Colab notebook). Autosaves every 10 real minutes and every simulated hour; resumes after interruption.

A local web app runs the queue overnight, autosaves every 10 minutes, tests each new ghost on 400 unseen levels and redraws these charts.

RESULTS

One night on a laptop

300 h

of simulated play: 16.2 million decisions

4.7 h

of real training time, CPU only

96%

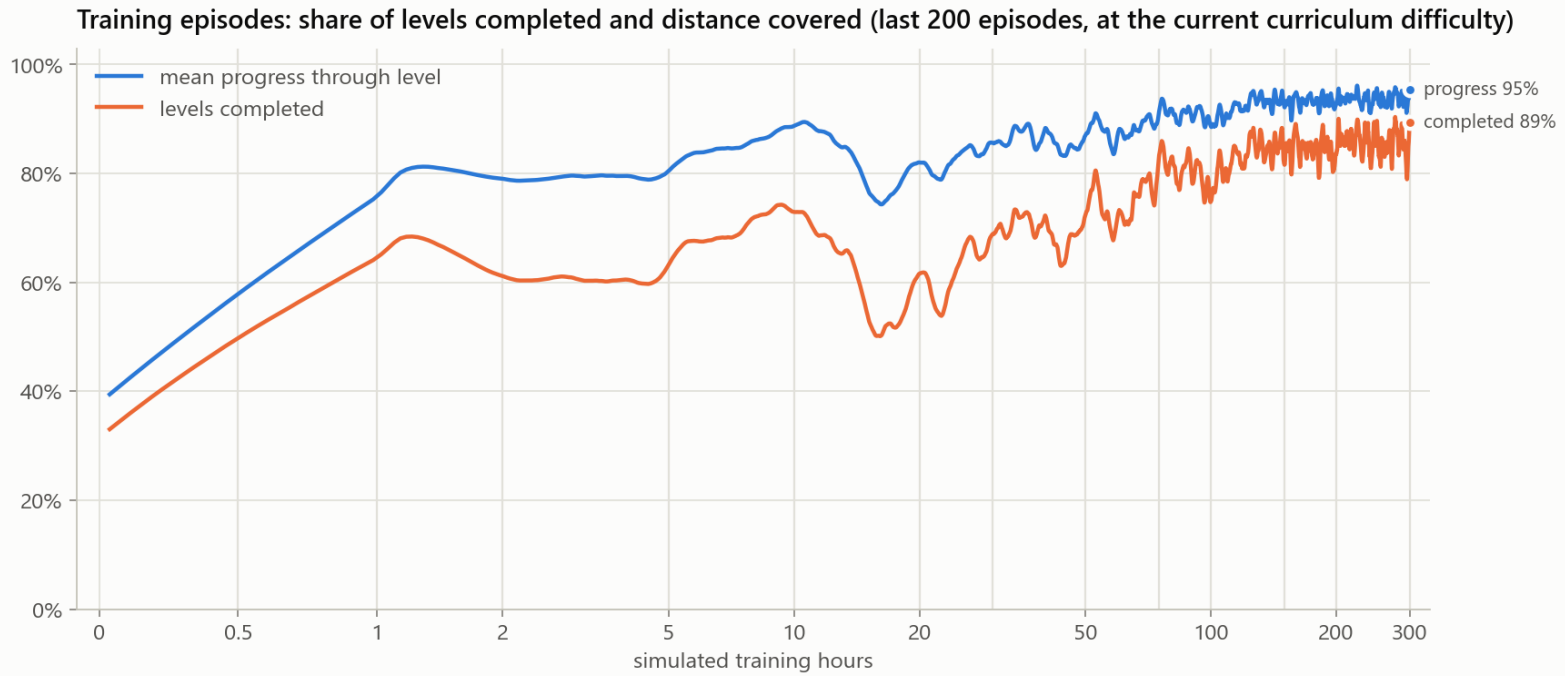
of never-seen levels completed at difficulty 0.75

85%

at the hardest training difficulty, 1.0

TRAINING

The learning curve



33% → 89%

of training levels completed, first rollout to last

Dips are good news

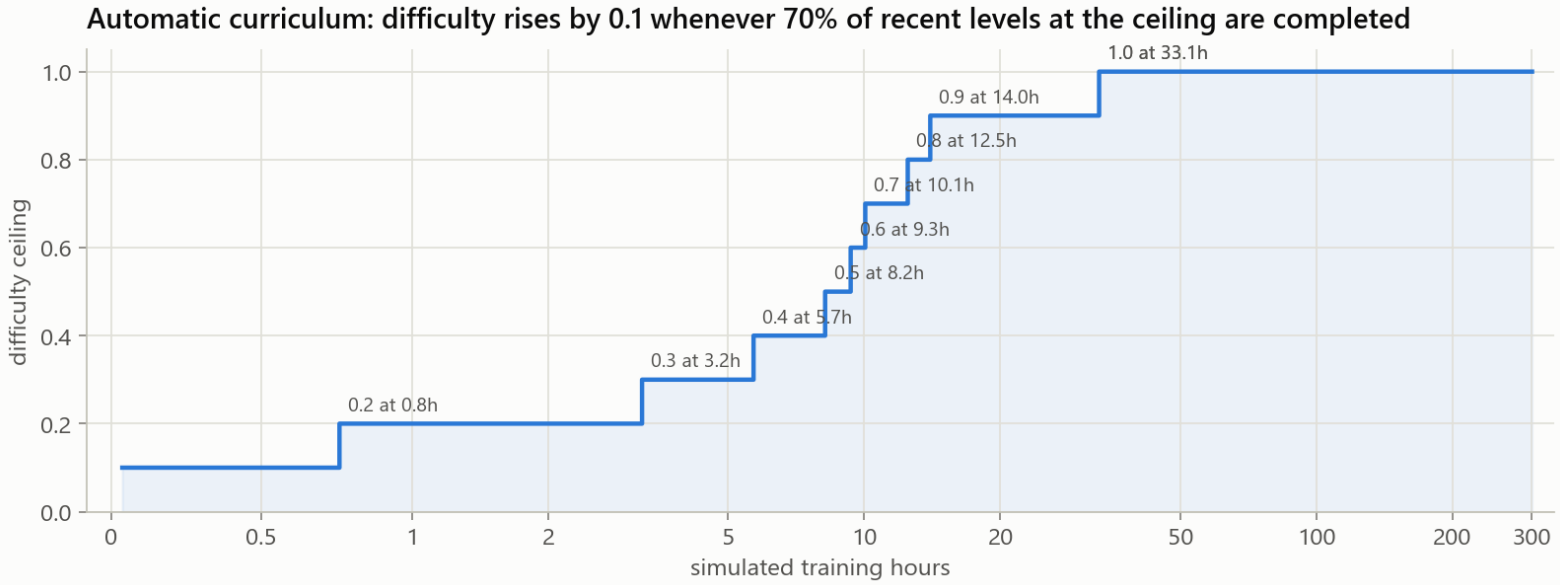
each one is the curriculum making the levels harder

Log time axis

most of the learning happens in the first 30 hours

TRAINING

A curriculum from easy to hard



0.1 → 0.9 in 14 h

eight promotions in the first 14 simulated hours

1.0 took until 33 h

the last step, with the widest gaps, was the hardest

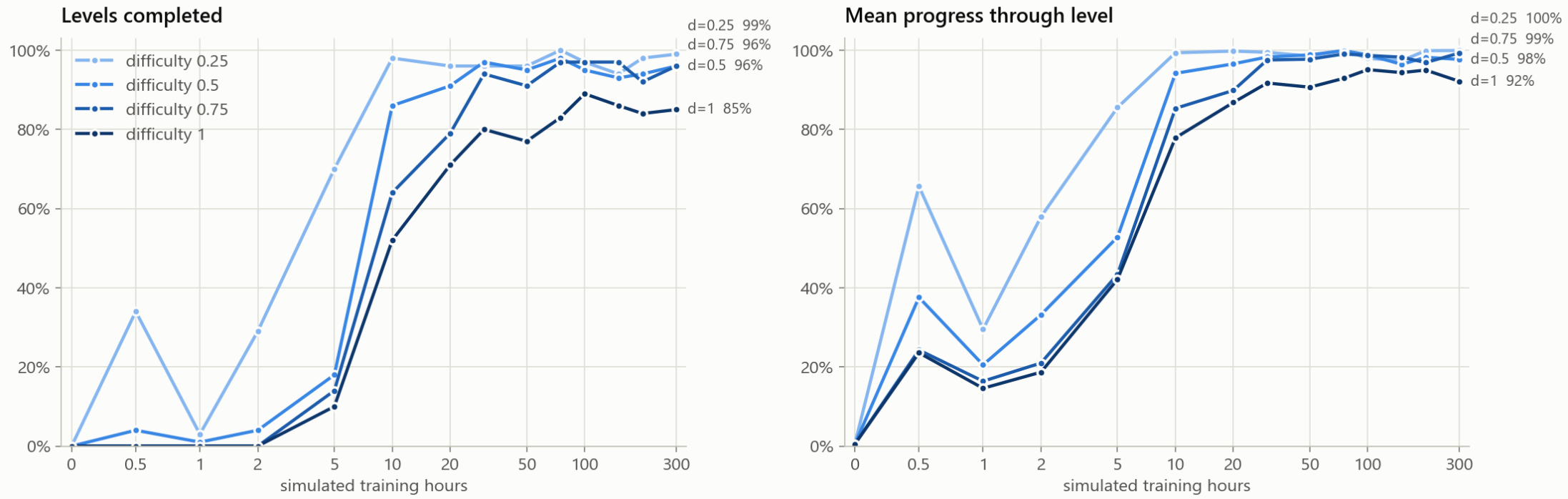
Promotion rule

complete 70% of the last 100 levels at the current ceiling

EVALUATION

Tested on levels it has never seen

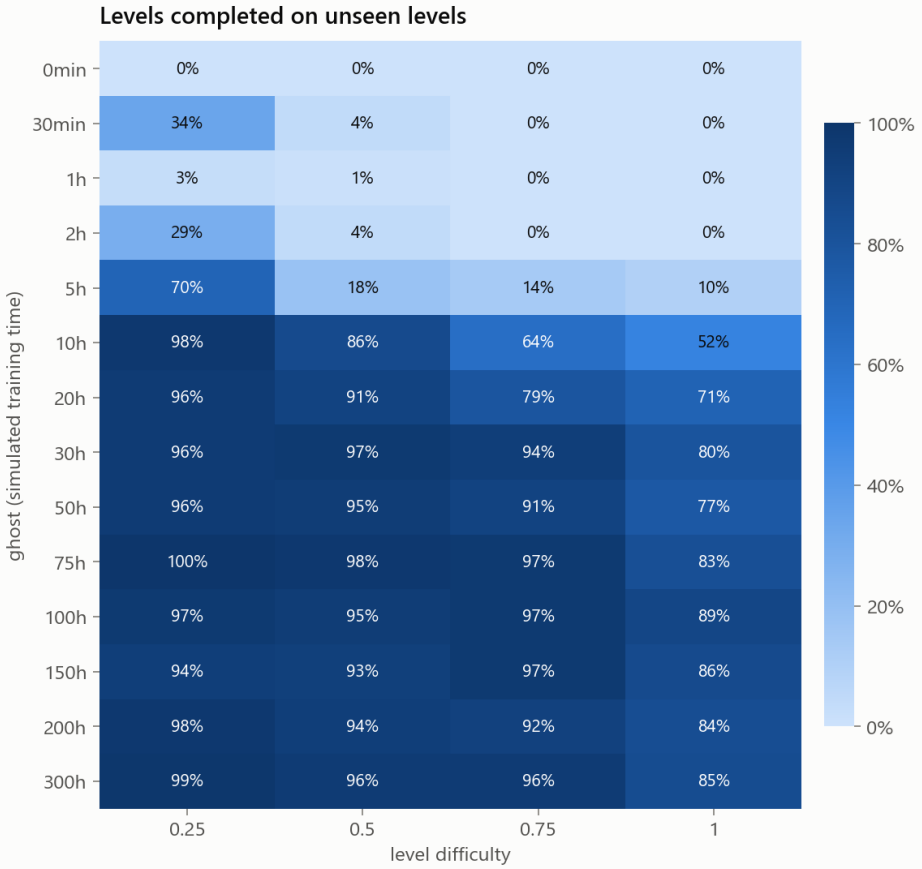
Every ghost tested on 100 never-seen levels per difficulty



Between 2 h and 10 h it goes from 4% to 86% at difficulty 0.5. From 30 h on it stays above 90% on all but the hardest levels.

EVALUATION

Every ghost at every difficulty



Read across: harder levels need more practice

At 10 h it finishes 98% of easy levels but only 52% of the hardest.

Read down: fast, then flat

From 30 h to 300 h, ten times more training, the hardest setting goes from 80% to 85%.

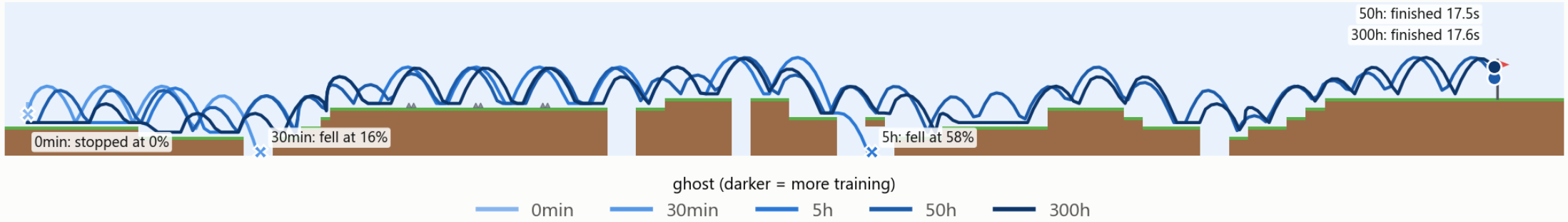
The top rows make the demo fun

Ghosts from 0 min to 2 h almost never finish a medium level: a human beats them easily.

GHOSTS

One unseen level, five stages of training

The same unseen level (seed 1000500, difficulty 0.6) played by ghosts from different training stages



0 min

Barely moves: 0%

30 min

Falls into the first pit at 16%

5 h

Falls into a later pit at 58%

50 h

Finishes in 17.5 s

300 h

Finishes in 17.6 s

Training buys survival, not speed: every ghost that finishes takes about 17 s.

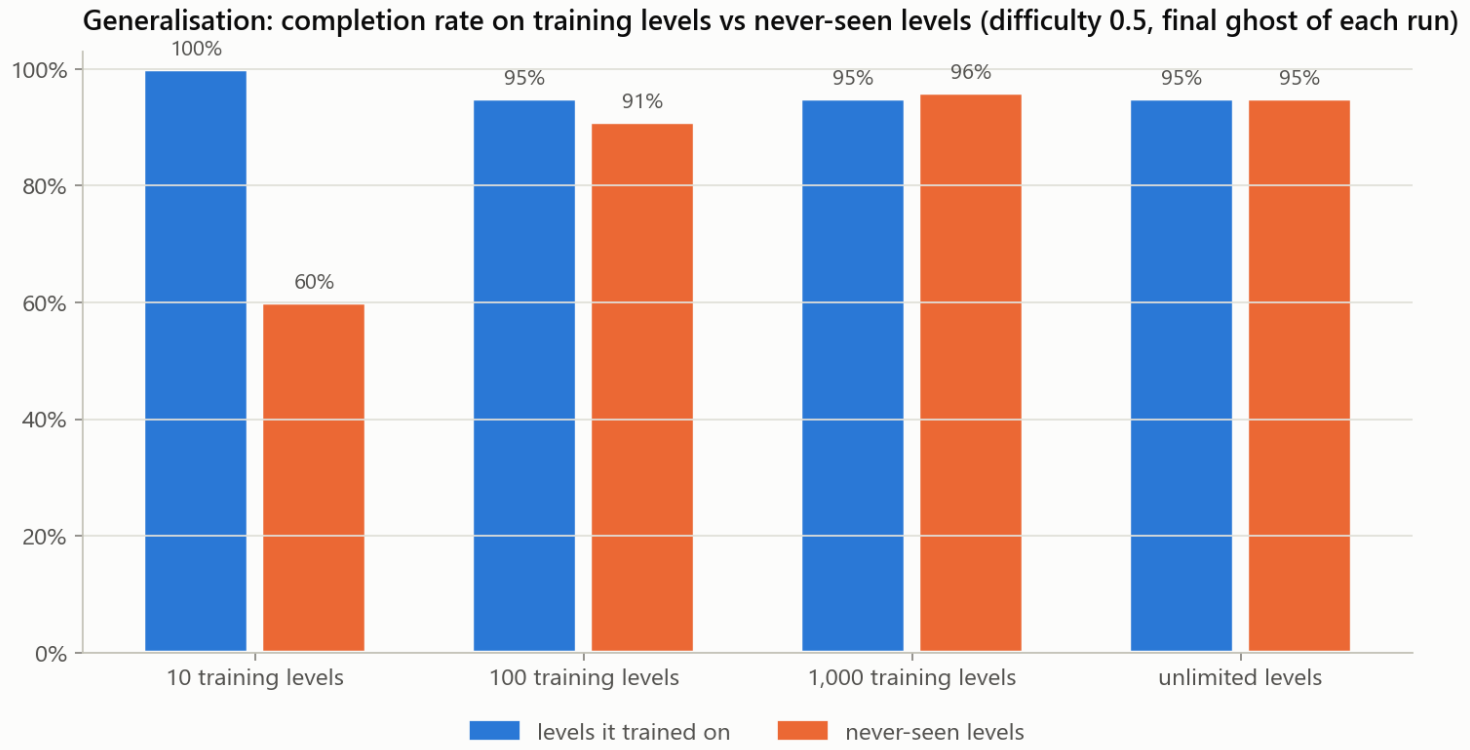
GENERALISATION EXPERIMENT

Trained on 10 levels: 100% on those, 60% on new ones

Same agent, same 100 simulated hours, same difficulty (0.5). The only change is how many different levels it practised on: 10, 100, 1,000 or unlimited.

GENERALISATION

More levels close the gap

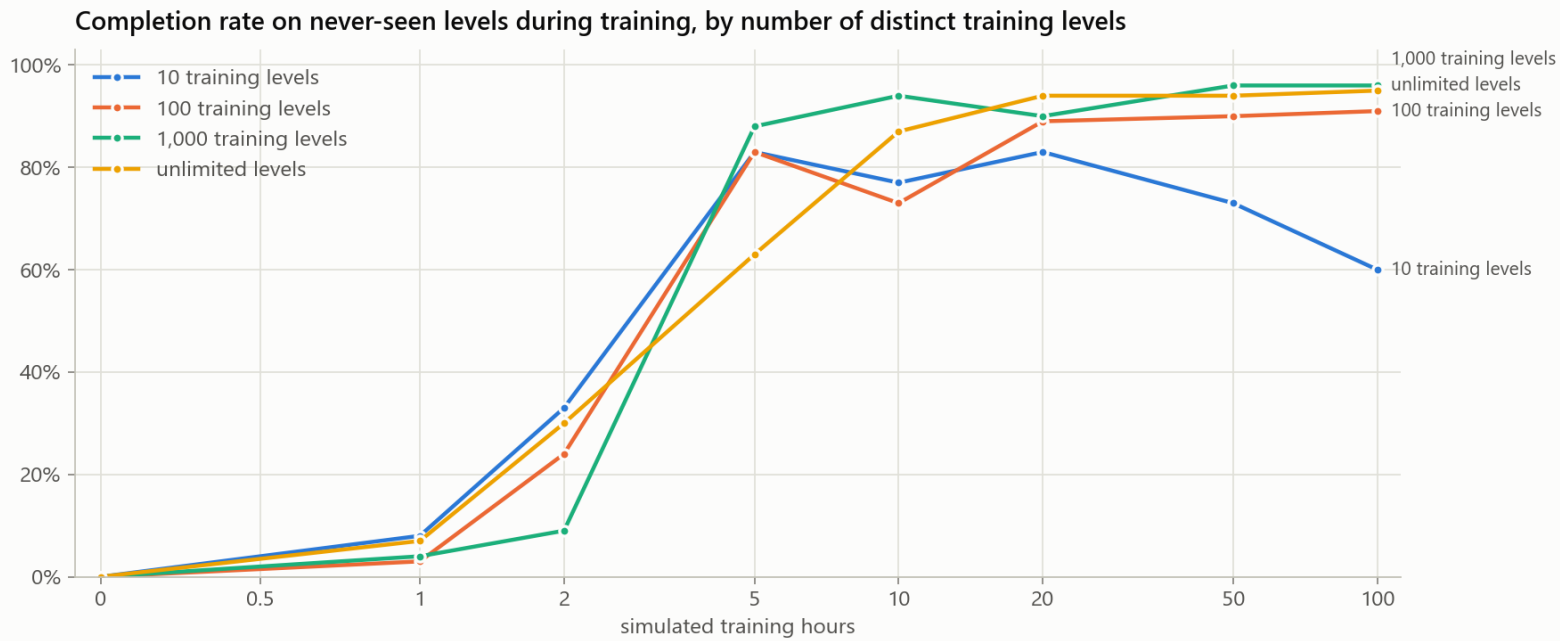


Levels	Own	New
10	100%	60%
100	95%	91%
1,000	95%	96%
unlimited	95%	95%

By 1,000 levels the gap is gone: the agent plays the game instead of replaying levels.

GENERALISATION

Memorising gets worse with time



10 levels: up, then down

83% on new levels at 5 h, 60% at 100 h. The longer it trains, the more it overfits.

Difficulty matters too

At 100 h, the curriculum agent finishes 89% of the hardest new levels; the unlimited agent trained only at 0.5 finishes 64%.

DEMO

Race the ghosts



- 14 see-through ghosts, 0 min to 300 h; you are the solid red one
- A fresh level every round, never used in training
- Compact leaderboard: arrows mark overtakes
- Arrows or A/D to run, Space to jump
- [and] change difficulty, up to 2.0

```
python demo.py
```

EXPERT LEVELS

Harder than anything in training



Gaps up to 6 tiles

was 4; the jump can clear 7

Spike runs of 4, often twice

plus spikes at gap edges and just after landings

One-tile pillars, less rest

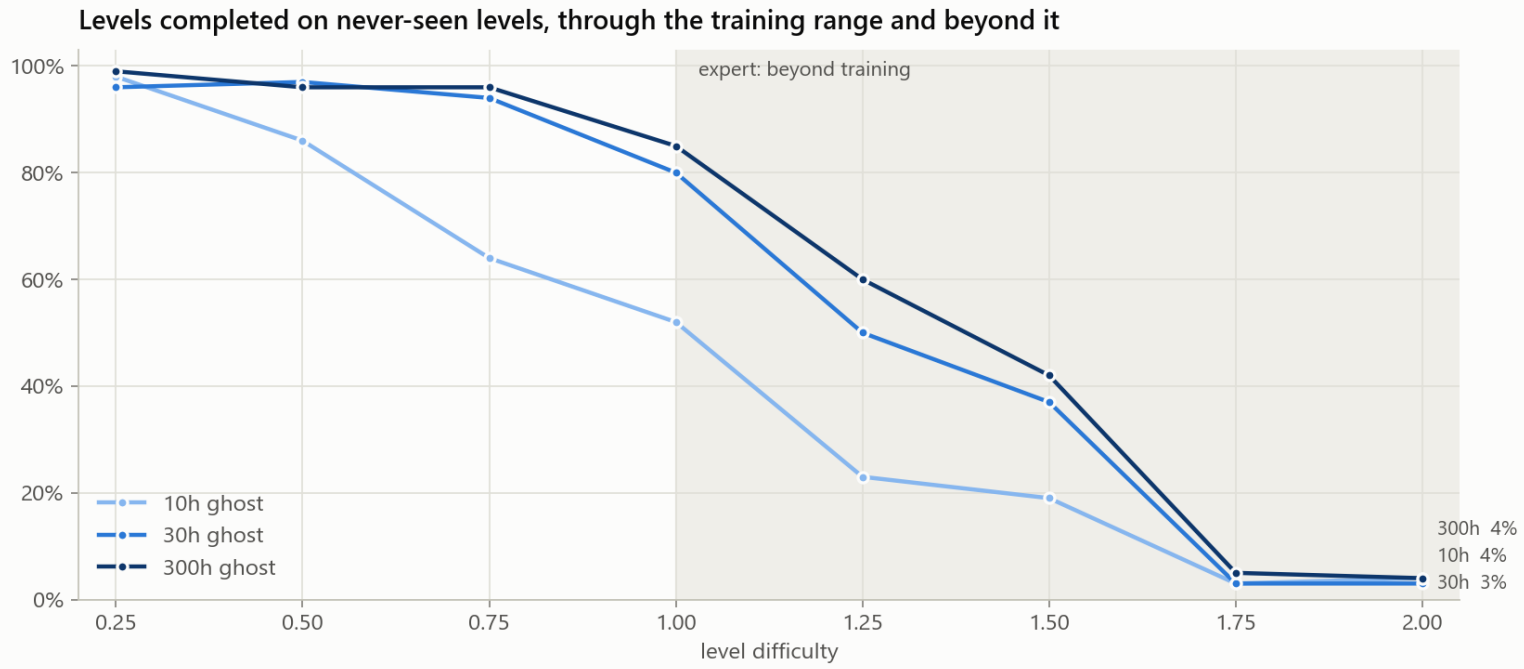
narrower platforms, shorter flat ground

Same tiles, same physics

no retraining; the solver checks every level is beatable

EXPERT LEVELS

Beyond training, even 300 h struggles



Difficulty	300 h ghost
1.0	85%
1.25	60%
1.5	42%
1.75	5%
2.0	4%

100 never-seen levels each. Now a good human can beat the best ghost.

WRAP-UP

What I learned, and what's next

What worked

- A tile view relative to the agent: cheap and level-independent
- A curriculum from easy to hard
- A new level every episode
- One shared game loop, so demo ghosts act exactly as in training

What was hard

- Guaranteeing every random level is beatable: a solver in the tests
- Reward for progress, so it learns before it ever reaches a flag
- Overnight runs: a full disk crashed two of them; autosave and resume recovered both

Next

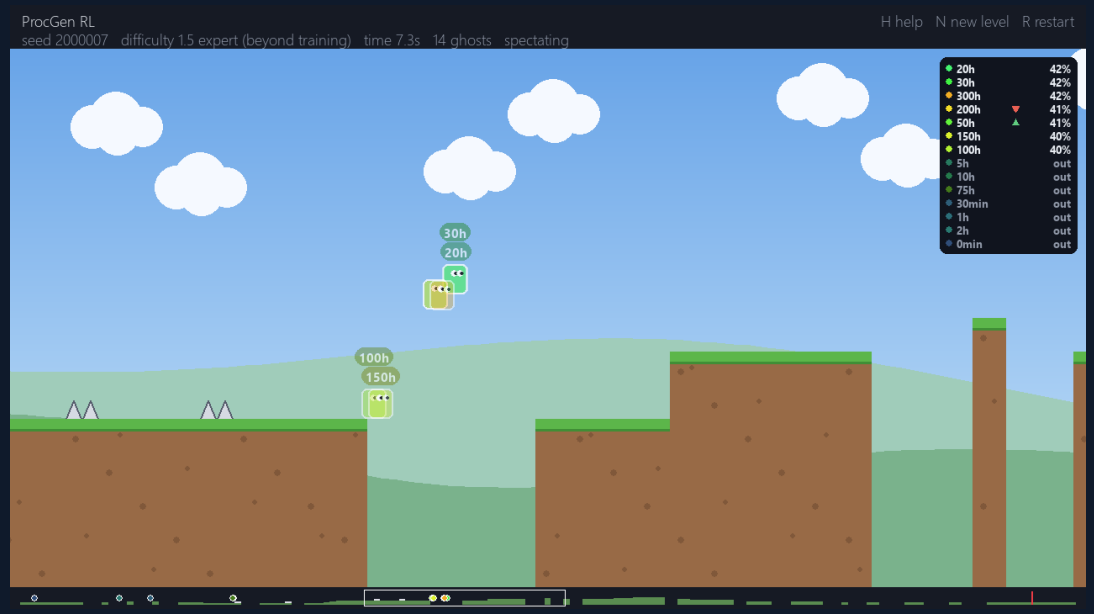
- Phase 2: enemies
- Pixel input with a convolutional network
- Record human runs and compare them with the ghosts

PROCGEN RL

Thank you

Questions, or want a race?

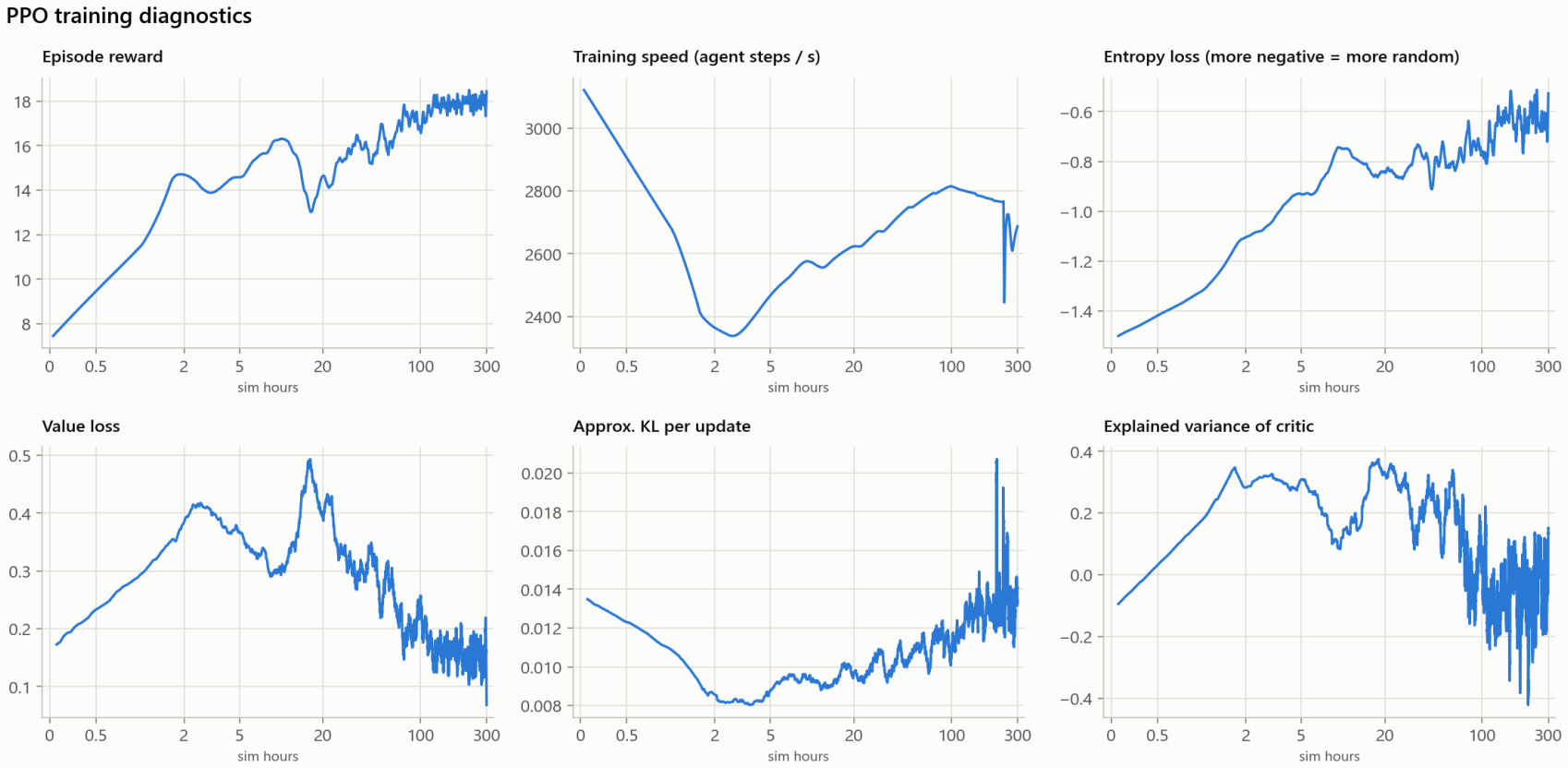
Code: github.com/DYNAMAXD/ProcGen_RL



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APPENDIX

PPO health checks



Reward rises from about 7 to 18 per episode.

Entropy falls slowly: fewer random actions.

KL stays near 0.01: stable updates.

Explained variance stays low, under 0.4: the critic is weak. A clear thing to improve.